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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Introduction to Machine Learning (course)



# Week 10: Assignment 10 (Non Graded)

Assignment not submitted

**Note : This assignment is only for practice purpose and it will not be counted towards the Final score.**

1) Which among the following is/are some of the assumptions made by the k-means algorithm (assuming Euclidean distance measure)? ***1 point***

- [ ] Clusters are spherical in shape
- [ ] Clusters are of similar sizes
- [ ] Data points in one cluster are well separated from data points of other clusters
- [ ] There is no wide variation in density among the data points

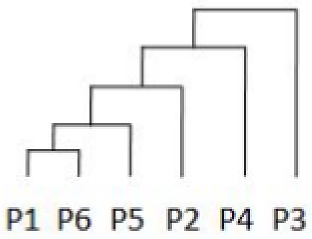
2) Consider the similarity matrix given below. ***1 point***

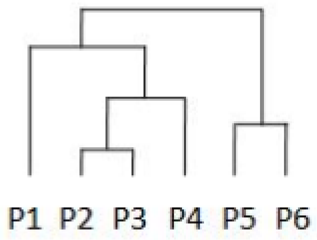
| | P1 | P2 | P3 | P4 | P5 | P6 |
|---|---|---|---|---|---|---|
| P1 | 1.00 | 0.70 | 0.65 | 0.40 | 0.20 | 0.05 |
| P2 | 0.70 | 1.00 | 0.95 | 0.70 | 0.50 | 0.35 |
| P3 | 0.65 | 0.95 | 1.00 | 0.75 | 0.55 | 0.40 |
| P4 | 0.40 | 0.70 | 0.75 | 1.00 | 0.80 | 0.65 |
| P5 | 0.20 | 0.50 | 0.55 | 0.80 | 1.00 | 0.85 |
| P6 | 0.05 | 0.35 | 0.40 | 0.65 | 0.85 | 1.00 |

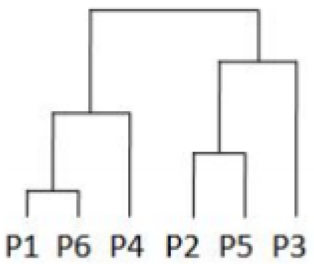
Show the hierarchy of clustering created by the single-link clustering algorithm.

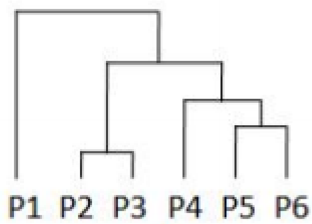
**Week 10 ()**

- Partitional Clustering (unit?unit=113&lesson=114)
- Hierarchical Clustering (unit?unit=113&lesson=115)
- The BIRCH Algorithm (unit?unit=113&lesson=116)
- The CURE Algorithm (unit?unit=113&lesson=117)
- Density Based Clustering (unit?unit=113&lesson=118)
- **Practice: Week 10: Assignment 10 (Non Graded) (assessment?name=273)**
- Week 10 Feedback Form : Introduction To Machine Learning (unit?unit=113&lesson=291)
- Week 10: Solution (unit?unit=113&lesson=229)
- Quiz: Week 10 : Assignment 10 (assessment?name=302)

- ◯ 

- ◯ 

- ◯ 

- ◯ 


3) In the CURE clustering algorithm, representative points of a cluster are moved a fraction of the distance between their original location and the centroid of the cluster. Would it make more sense to move them all a fixed distance towards the centroid instead? Why or why not? ***1 point***

- ☐ Yes, because this approach will ensure that the original cluster shape is preserved.
- ☐ No, because this approach will not be as effective against outliers as the original approach.

4) Suppose while performing DBSCAN we randomly choose a point which has less than MinPts number of points in its neighbourhood. Which among the following is true for such a point? ***1 point***

- ◯ It is treated as noise, and not considered further in the algorithm
- ◯ It becomes part of its own cluster
- ◯ Depending upon other points, it may later turn out to be a core point
- ◯ Depending upon other points, it may be density connected to other points

**Week 11 ()**

**Week 12 ()**

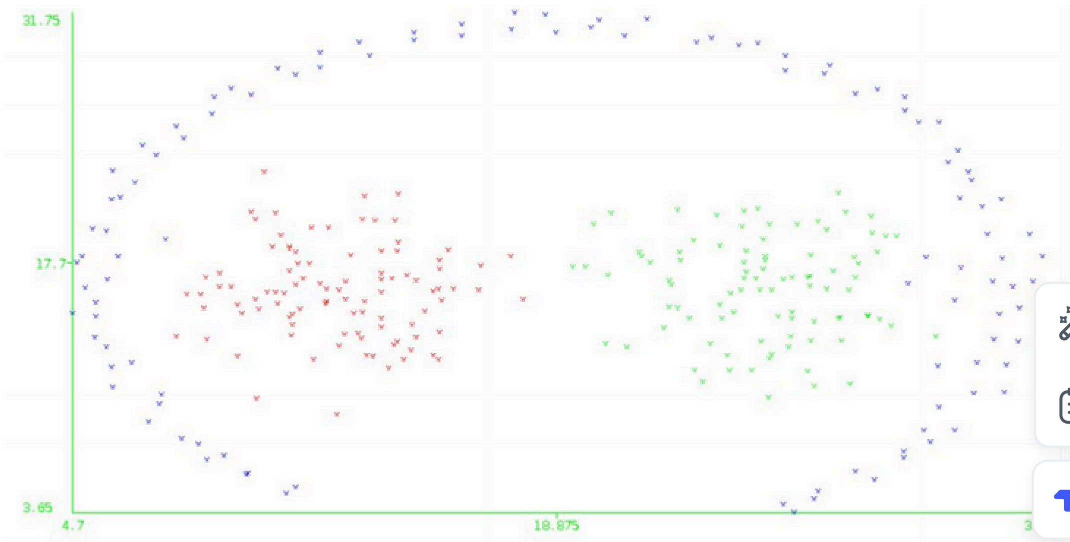
**Text Transcripts ()**

**Download Videos ()**

**Books ()**

**Problem Solving Session - July 2024 ()**

5) Consider the following image showing data points belonging to three different clusters (indi- cated by the colours of the points). Which among the following clustering algorithms will perform well in accurately clustering the given data? ***1 point***




- K-means
- Single-link hierarchical
- Complete-link hierarchical
- DBSCAN

**Check Answers and Submit**